Lab: Blender Materials

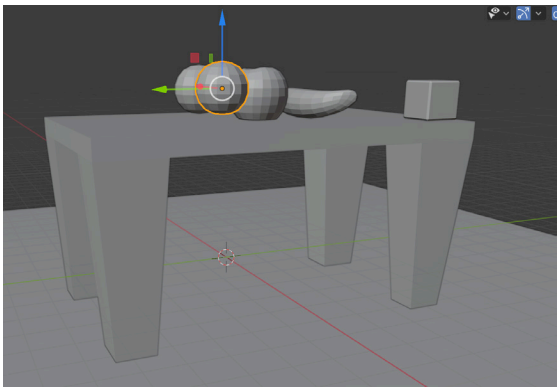
At this point we know how to model a basic mesh, whether it is a spaceship, throne or treasure chest. The next step in understanding 3D graphics is what is a 3D material.

In the real world many objects have the same shape, but are made of different materials. For instance a bowling ball and foam Nerf ball are the same shape, they are both spheres; but while the Nerf ball is made a soft foam material, the bowling ball is a made of a thick heavy poly-urethane material.

Another example would be a table. The table could be made of wood, it could also be made of thick plastic like patio furniture. It could even be a fancy metal table like those found inside restaurant kitchens. In all of these case the shape is the same, but the material is different.

3D programs like Blender allow us to create a 3D material to apply to a 3D mesh. The material allows us to set properties of the material including color, light reflectivity and other properties.

This lab will go over the basics of creating and editing a material in Blender.

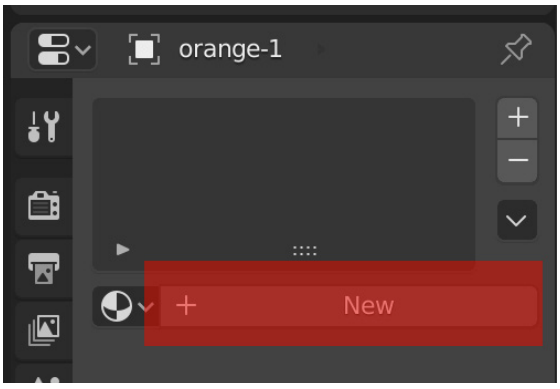


Setup

Begin by loading *materials.blend* into Blender. Inside this file you'll see a table. On the table is a bowl of fruit containing an apple, orange and banana inside a bowl.

Feel free to orbit, pan and zoom around the scene.

Create a Material



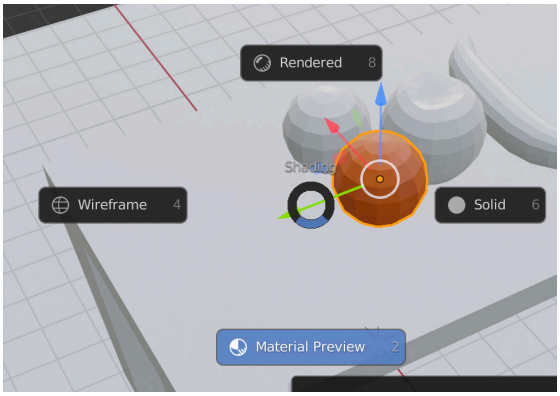
This scene is good, but could use some materials to make it more realistic and give the scene some character.

Begin by selecting *Orange-1* in the scene. In the *properties* panel click the materials icon, which looks like a checkered ball. This is where we will add a material to our orange.

In the materials tab click the option marked '*New*' to create a new material. By default it will be called something like *material.001* or *material.002*, we can rename it by double-clicking on the name and changing it. Let's call it '*orange material*.'

When you created a material you will have access to many properties that describe how the material will behave. These

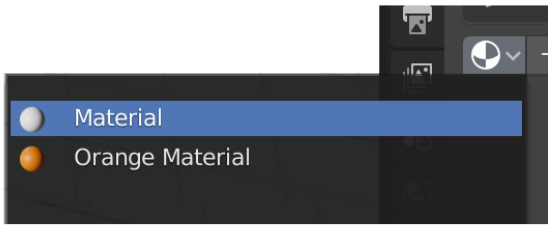
include things such as the base color of the material, how shiny or how the material reacts to light sources, and other related properties. For now, let's just focus on the color of the material, also known as the base color.



For the orange material set the base color to the color orange. If you like to see a preview of the material open up the preview section of the materials tab.

You orange material should now be applied to the orange mesh in the scene. You can check this by viewing the Materials Preview in Blender. You can click on the *Materials Preview* button in the upper right hand side of the 3D Viewport or press the '*z*' key and choose *Material Preview* from the options.

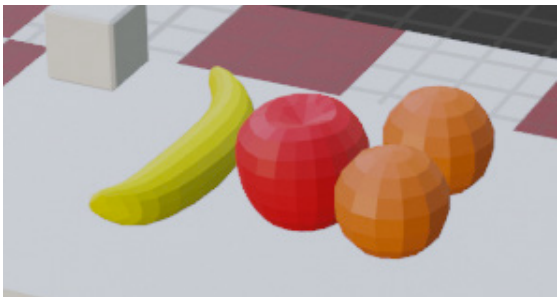
Apply the Material to a Different Mesh



Now that we createdthe orange material let's apply it to *Orange-2*. Begin by selecting *Orange-2*. Go to the materials tab and choose the 3D checked ball on the left side of the tab. This will list all of the materials you have created in the scene.

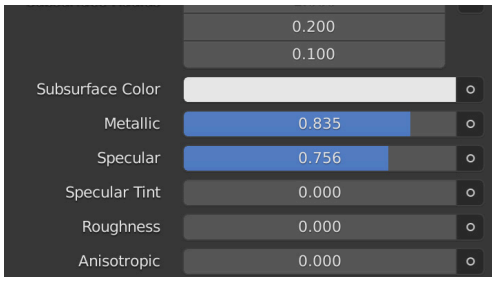
Choose the existing orange material to apply it to *Orange-2*

Create Materials for Apple and Banana

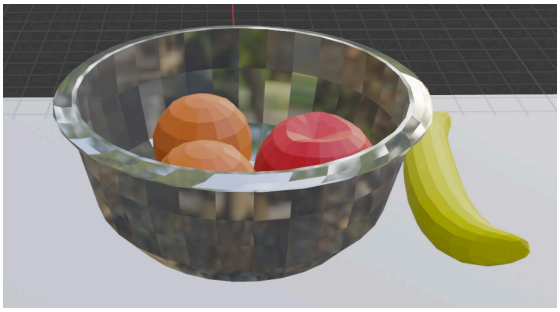


Now that we have an understanding the basics for creating, applying and previewing 3D materials in Blender let's create an Apple Material and Banana Material. Give the Apple Material a base color or red and apply to the Apple Mesh. Give the Banana Material a base color of yellow and apply it the Banana Mesh.

Create a Shiny Metal Texture for the Bowl



Let's delve a bit deeper into material properties. One major aspect of a property is how well it reflects light. Think about the difference between shiny metal and burlap. Metal reflect more more light than burlap would.



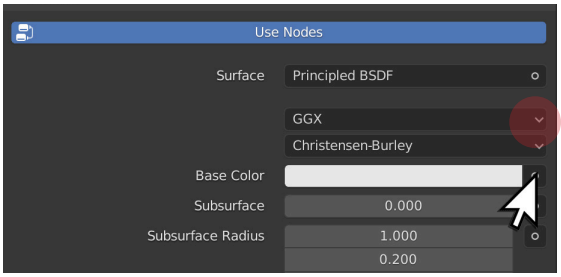
Select the bowl and create a material to it called Metal Material. Set the base color to a light gray. Now go to the property marked '*Roughness*'. Roughness means how much light will be reflected from the material, it goes 0-1. Zero means it will reflect A LOT of light, 1 means almost no light would be reflected. Metal would be a low number like 0 or 0.1, the burlap would be a high number like 0.9-1. Set the roughness of the metal material to 0.

Next we'll play around with the metallic property. Metallic

property is for how much of the environment will be reflected in the material. Think about a mirror, that would reflect almost of the scene. A shiny car hood would reflect most of the scene. A shiny spoon would reflect some, but not all of the scene. This will be a shiny metal bowl, so set the Metallic property to 1.

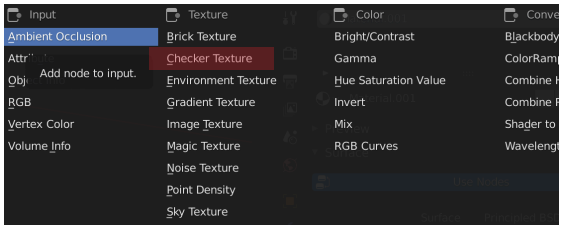
Switch to Materials Preview and you should see the light bouncing off the bowl and the environment reflecting in it.

Material Textures

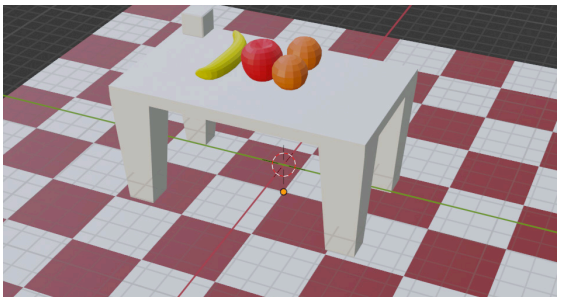


In addition to having a base color you can also apply a texture to your 3D mesh. A texture is a complex surface for an item. A texture could a brick texture, checked texture or an image you bring into Blender. Blender has built-in textures as well as the ability to bring in your own texture images.

In this lab we will created a checked texture for the floor in our scene. Begin by selecting the floor mesh. Then create a new material and label it '*Floor Material*'.



Instead of choosing a base color, click the white color on the right side of the base color option. This will open up a dialogue box like the one on the right. Choose the option called '*Checker Texture*.'



After applying the checkered texture you'll see a black and white checkered pattern appear on the floor mesh in Blender. This checkered texture built into Blender and can be modified. In the materials tab you can change the colors of the checked patterns using *color1* & *color2* options. You can also scale the size of the checkered pattern as well using the scale option.

Change the checkered colors to two new colors and adjust the scale to your liking.

File Texture Material

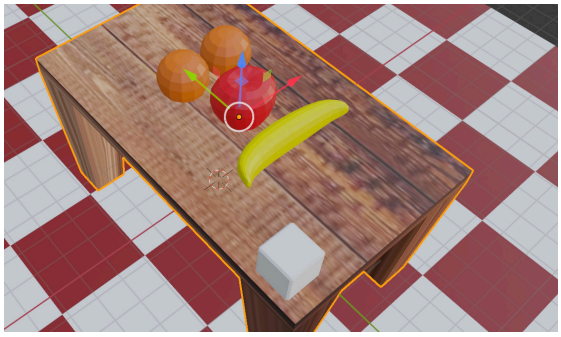


Blender has built-in textures such as checkered textures and brick textures, and it also allows you to apply your own textures in the way of a file image.

Go to the textures folder in the lab folder and you'll see an image file called '*Wood-Texture.jpg*'. In this lab we are going to apply this wood texture to table.

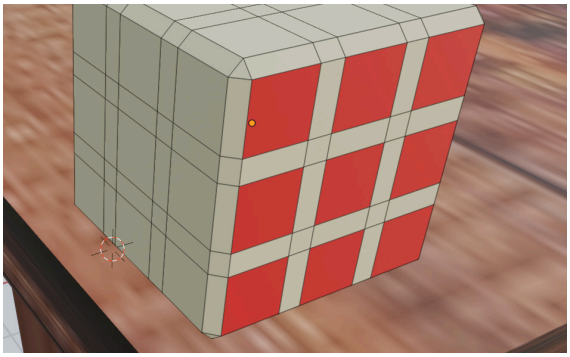
Select the table 3D mesh inside Blender. Create a new material and call it '*Wood Material*'. Go to the base color tab and choose the white circle on the right side of the option to open up the dialogue box. Choose the option marked '*File Texture*,' choosing

this will open up a file picker for Blender. Find and choose '*Wood-Texture.jpg*'. This will apply the wood texture image to the table.



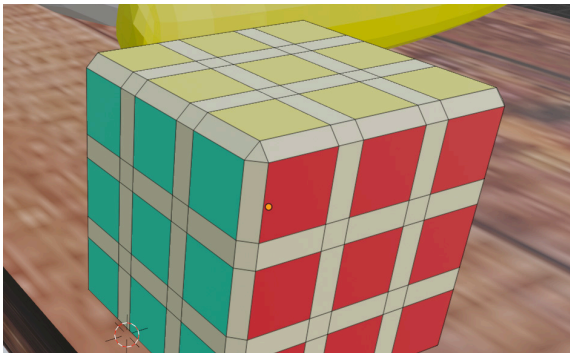
If you want to change the texture to a different image choose the 'Open File' icon in the materials menu, then choose a new image to apply. Feel free to try applying different textures in the lab folder to the table to see different results.

Apply Multiple Materials



We've discussed different ways to create materials in this lab. Some objects may need to have more than one texture. For instance if you were modeling a hammer you would want a wood texture for the handle of the handle and also a metal texture for the top part of the hammer. If you had a model of a home would might new a material for the stucco exterior, but also the glass applied to the windows.

On the table in our scene we have a rubik cube. If you have ever seen a rubik cube then you know that rubik cube have multiple colors as part of the design. Let's try to add multiple texture to the rubik cube in our scene.

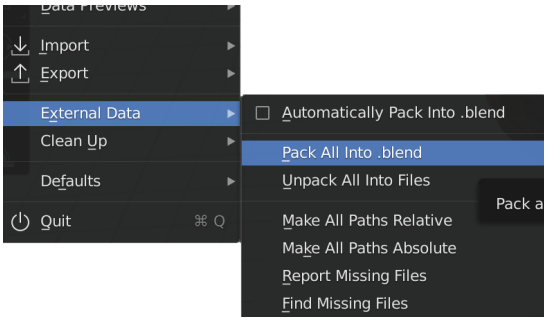


Begin by selecting the rubik cube. Create a new material and call it *rubix-1*. Give the *rubix-1* material the base color of your choice.

With the rubik mesh selected switch from *Object Mode* to *Edit Mode*. Select the faces on one side of the rubik cube and choose your rubix-1 material, then choose the assign option in the materials tab. You should see your material apply but only to the one side of the mesh.

With the rubik mesh selected create a second material and call it *rubix-2*. Give it the base color of your choice. Apply *rubix-2* to another side of the rubik cube. Repeat the process again and create and apply a third material to the rubik cube. When you preview the scene the rubik cube should have multiple materials being applied.

Managing Texture Files



When you start to use texture files you are beginning to use additional external documents in addition to our *.blend* file. To make sure that your textures come through properly keep your texture images in a folder along with the blend file.

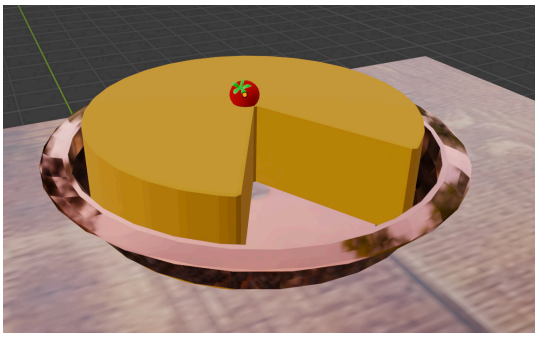
Another way to manage your textures would be to load into the blend file itself. To do this go to *File > External Data > Pack All Into .blend*. Once you save all of your image files will be embeded into the .blend file. If you would like this to be the default of the program set the *Automatically Pack Into .blend* option under *File > External Data*.

Submit

Great job! This should give you an insight of how to create and apply materials to meshes in Blender.

Save and submit your file as *lastname-materials.blend*. If you kept the textures as external files put them into a folder with the *.blend* file and zip and submit the folder.

Extra Credit (+2 points) Not Required



Open up your *Modifiers.blend* scene and add materials for the:

- table
- cake
- dish
- strawberry